TrTr-CMR: Cross-Modal Reasoning Dual Transformer for Remote Sensing Image Captioning

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***Abstract*— Remote sensing image captioning (RSIC) is an interesting but challenging cross-modal reasoning task for computer vision and natural language processing. Most of the recent popular approaches for RSIC utilize encoder–decoder architectures, which focus on visual features captured by convolutional neural network (CNN)-based encoder and semantic information by recurrent neural network (RNN)-based or long short-term memory (LSTM)-based decoder, but encounter difficulties with multiscale, multicategories, and direction ambiguity challenges. To make the most of semantic understanding ability of Transformers, in this article, we propose a new attention-based visual-linguistic reasoning framework with dual Transformer for RSIC. Specifically, Swin Transformer (SwinT) encoder with shifted window partitioning scheme is introduced for multiscale visual feature extraction to discover the intrinsic relationship in the objects, and then, a Transformer language model (TLM) with self-attention and cross attention is designed as the decoder to generate a well-formed sentence for the image. Extensive experiments are conducted on the public RSIC benchmark datasets, including UCM-Captions, Sydney-Captions, and RSICD. The impressive performance verifies the effectiveness and superiority of the proposed method. In addition, the source code and models of this work are publicly available at *https://github.com/LianYi233/TrTr-CMR*.**

***Index Terms*— Image captioning, natural language processing, remote sensing, Transformer.**

I. Introduction

CURRENT progress in high-resolution optical imaging and aerospace techniques makes detecting and analyzing the Earth's surface easier and more convenient, which paves the way to remarkable works in remote sensing scene parsing [1], semantic segmentation [2], [3], image registration [4] [5], salient object detection [6], and moving object tracking [7], [8]. However, the aforementioned works mainly deal with the attributes of single modal of images but ignore the relationship of the image content.

Motivated by vision-to-language (V2L) tasks in natural language processing, image captioning [9], [10], visual question answering (VQA) [11], and text-to-image synthesis [12], [13] have attracted an increasing attention in the field of optical remote sensing imagery in recent years. As the fundamental V2L task among them, remote sensing image captioning (RSIC) aims to generate a human understandable text-based description of a given remote sensing image, which exactly describes the objects and high-level relationships in the remote sensing images. Leveraging the powerful feature extraction ability of convolutional neural networks (CNNs), Qu et al. [9] released the first benchmark dataset of RSIC task and proposed a novel multimodal neural network framework, which combines the visual feature from CNN and textual information from recurrent neural network (RNN) for image captioning generation. Lu et al. [10] explored the large-scale RSIC dataset collecting aerial images from various data sources, such as Google Earth, Tianditu, Baidu Map, and so on.

Comparatively, generating accurate and flexible sentences for remote sensing images is more challenging than that for natural images. The main difficulties are as follows.

1) *Scale Ambiguity*: Remote sensing images are obtained from UAVs or satellites. Due to the height of camera, the ground object sometimes indicates different semantic information under different scales.
2) *Category Ambiguity*: The ground objects with similar appearance in different regions may represent the different landcover or land use categories; in a single region, it would contain multiple categories of objects occasionally.
3) *Direction Ambiguity*: The remote sensing images are always captured in the "view of God," which means there is no concept about top/bottom or left/right, and never exist a fixed observing direction. This brings difficulty in depicting the relative relationship between the objects.

Many researchers have studied image captioning approaches to describe the content of remote sensing images by concise

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and precise sentences. Li et al. [11] introduced attention mechanism into V2L, including text-guided attention and semantic-guided attention, to explore the mapping between the visual features and the semantic texts. Wu et al. [14] designed an end-to-end captioning architecture utilizing CNN to extract global visual feature and context vector calculated by scene attention from long short-term memory (LSTM) network to generate a smooth sentence. Wang et al. [15] proposed a two-stage captioning framework, which consists of a CNN-based word extractor and a Transformer-based sentence generator. To alleviate the multiscale ambiguity in remote sensing images, GLCM [16] learned the relationship between the words and the local visual features by global–local feature encoding network. Then, the self-attention decoder could break the limitation of time step in LSTM and output more flexible words. In addition, a novel meta learning captioning paradigm is explored in [17] with the help of the transferred knowledge from a well pretrained classification model.

So far, the mainstream architecture for RSIC is the encoder–decoder framework. The CNN encoder extracts the geographic information from the image; then, the visual features are decoded into natural language sentences by RNN-based or LSTM-based decoder. To address the multiscale problems in remote sensing scenarios, MLAT [18] utilized ResNet50 as the encoder and designed a Transformer decoder with an LSTM as the connectivity between Transformer encoding and decoding layers, which aggregated the features from each encoder. Compared with traditional CNN-based encoders, the vision Transformer (ViT) with shifted window partitioning could better discover the intrinsic relationship in the image. Moreover, the Transformer language model (TLM) has a more promising ability and fewer grammatical errors in sentence generation, and higher training efficiency than LSTM networks. The blockbuster success of the Transformer models [19] has verified that self-attention and cross-attention structure are suitable for both vision and language tasks. Therefore, inspired by recent advances in Transformer architecture, we introduce a dual Transformer model into our encoder–decoder RSIC framework for a more powerful semantic understanding ability.

In summary, the main contributions of the proposed method could be summarized as the following fourfold.

1) We propose a dual Transformer structure cross-modal reasoning framework for RSIC, which has more powerful potential for recognizing and understanding the ground objects and their relationships in the image.
2) In the encoding stage, the SwinT encoder with a shifted window partitioning scheme utilizes alternative window-based multihead attention and shifted window-based multihead attention for effective visual feature extraction.
3) In the decoding stage, an attention-based TLM is adopted as a sentence generator. The subencoder and subdecoder that consist of self-attention and cross attention are employed for high-performing sentence generation.
4) The extensive experiments are conducted on three public RSIC benchmark datasets, including UCM-Captions, Sydney-Captions, and RSICD. Our method achieves superior performance compared with 11 state-of-the-art RSIC approaches on the three datasets and generates well-formed sentences for the remote sensing images.

II. Related Work

A. Image Captioning

Image captioning, which is also known as vision-language generation, is one of the most imperative multimodal tasks in computer vision and natural language processing. The goal is to transform an image into a sequence of words according to the objects and their relationship.

The early proposed image captioning methods usually take retrieval-based [20], [21] [22] or template filling [23], [24] approaches. Farhadi et al. [20] evaluated the distance of image space and corpus by mapping the k-nearest neighbors to the space of meanings with a triplet of {object, action, scene}. Im2Text [21] presented a captioning generation method that retrieving the most relevant images from a large-scale captioned photograph database, reranked the candidate images according to image contents, and aggregated the sentence via captions of top-ranking retrieved images. Karpathy et al. [22] designed a bidirectional image–sentence retrieval approach by embedding fine-level image fragments and sentence fragments into a common space. *BabyTalk* [23] introduced conditional random field (CRF) to process candidate objects from the image and transformed the prediction keywords into coherent sentence using n-grams language model. *TreeTalk* [24] detected the visual feature similarity to discover useful words (i.e., tree fragments) and generated high-level linguistic expressions utilizing these fragments based on templates or rules.

The typical retrieval-based methods usually could not describe the contents, which are unfamiliar with searching dataset. For language models in template-based methods, it is challenging to learn the linguistic constraints according to common sense and grammatical rules. Sometimes, the predicted sentences are not as fluent as human writings. Therefore, the learning-based encoder–decoder paradigm has become the mainstream architecture for image captioning with the advent of deep learning. The framework is end-to-end trainable by maximizing the likelihood probability of reference sentences. Inspired by related works in machine translation task, *show and tell* [25] made a quite simple recipe of neural image caption, which combine a vision CNN pretrained on image classification task and a language RNN to translate the image into a sentence. Thereafter, *show, attend, and tell* [26] introduced two variants of attention mechanism to connect CNN encoder and LSTM decoder, which are hard stochastic attention and soft deterministic attention, to focus on the salient regions. *Show, recall, and tell* [27] imitated the behavior of human recalling to the image captioning task and designed a text-retrieval model to make the recalled words more relevant to the image. SCA-CNN [28] yielded the modulated CNN features through spatialwise and channelwise attention in multilayers. Ji et al. [29] aggregated global and local information among layers, which enhances the performance of LSTM with global-enhanced attention.

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Recently, Transformer networks have been widely applied in place of traditional convolution architectures. On this line, several variants of Transformer have been adopted to improve visual feature representation or language generation models. The first convolution-free captioning network CPTR [30] is explored with a ViT model [31]. BERT-like decoder [32] fused visual and textual modalities with the advantage of pretraining from external corpora. Benefitting from the striking ability of understanding and reasoning of modern generative pretraining (GPT) models, PromptCap [33] designed a question-aware captioning approach with natural language prompts via GPT-3 [34] in-context learning.

B. Remote Sensing Cross-Modal Reasoning

Similar to natural V2L tasks, there are several hot issues of remote sensing cross-modal reasoning tasks, including RSIC [10], RSIC captioning (RSICC) [35], remote sensing visual question answering (RSVQA) [36], and remote sensing visual grounding (RSVG) [37]. RSIC is the fundamental task of generating the description of a remote sensing image, while RSICC aims at describing the land cover changes in multitemporal remote sensing images [38] [39]. In this article, we focus on the RSIC task.

Most existing RSIC approaches adopt the popular CNN and LSTM encoder–decoder architecture. Shi and Zou [40] first investigated the text generation problem for Google Earth and GaoFen-2 satellite imagery and built a simple fully convolution network (FCN) based on VGG-Net [41]. In [10], many visual feature representation approaches are employed as encoder, including CNN deep features and handcrafted features, such as scale-invariant feature transform (SIFT) [42], bag-of-visual words (BOV) [43], Fisher vector (FV) [44], and vector of locally aggregated descriptor (VLAD) [45] followed by the sentence generator RNN or LSTM. To alleviate the drawback of scale ambiguity, [46] designed a multiscale cropping strategy during the training stage of CNN encoder. Huang et al. [47] further improved this multiscale feature representation by concatenating with denoising features in spatialwise and channelwise. To overcome the overfitting in the CNN-LSTM framework caused by synonyms phenomenon in the traditional training stage, Li et al. [48] proposed a novel truncation cross-entropy (TCE) loss in place of conventional cross-entropy loss, aiming at alleviating the overfitting problem, which helps generate more concise sentences. Zhao et al. [49] utilized the pixel-level classification result of the remote sensing image as an auxiliary of high-level feature representation. Li et al. [50] designed a multilayer tri-LSTM decoder structure, which is composed of vision LSTM, content LSTM, and output LSTM.

Although CNN-LSTM framework is a commonly used and sound approach, recent studies [18], [51], [52], [53], [54], [55] have introduced the advanced Transformer architecture for RSIC to enhance the quality of feature extraction or sentence generation. Unlike LSTM, a Transformer is a stack of identical layers primarily composed of self-attention and without time-dependent structures, so the training efficiency is often higher than LSTM decoders. Shen et al. [51] designed a variational autoencoder-based two-stage multitask learning model, which utilized a Transformer text generator instead of LSTM. Zhuang et al. [52] enhanced the grid features extracted from CNN by a Transformer model. Yang et al. [53] proposed a lightweight Fourier Transformer with image–text contrastive learning for visual and textual feature alignment. Meng et al. [54] proposed a prior knowledge-guided Transformer (PKG-Transformer), which first extracts scene-level features by ResNet [56] and object-level features by Faster R-CNN [57]. Then, a Transformer decoder integrated with prior knowledge augmented attention provided the knowledge that promotes sentence generation. Afterward, MG-Transformer [55] introduced a pretrained CLIP model to achieve comprehensive image representation. The Transformer decoder established the relationship between text and image by meshed cross-attention mechanism.

III. Cross-Modal Reasoning Framework

A. Cross-Modal Reasoning Architecture

The overall architecture of the proposed method is shown in Fig. 1. We conduct the encoder–decoder architecture as the overall framework, which could be trained in an end-to-end manner. The entire model consists of two subnetworks, namely, the SwinT encoder and the TLM. The input remote sensing image is cropped into nonoverlapped windows to conduct window-based or shift window-based attention for visual feature extraction. In the decoding subnetwork, the TLM decodes the features into a readable sentence that describes the image content.

B. Swin Transformer Blocks

SwinT block cropped the original image (or feature map) into several nonoverlapped windows, where each window consists of $M \times M$ patches. These patches will serve as the input tokens of the SwinT. To better discover the intrinsic relationship between the windows, the shifted window partitioning scheme is adopted as a complementary of the regular window partitioning, as shown in Fig. 2. Specifically, for the input feature map X^l in layer l, the output of the SwinT block is calculated in a residual manner based on two alternate used multihead self attention (MSA) modules followed by a multilayer perceptron (MLP), which could be formulated as follows:

$$\begin{aligned}
\hat{X}^l &= \mathbf{X}^{l-1} + \text{W-MSA}\left(\text{LayerNorm}\left(\mathbf{X}^{l-1}\right)\right) \\
\mathbf{X}^l &= \hat{X}^l + \text{MLP}\left(\text{LayerNorm}\left(\hat{X}^l\right)\right) \\
\hat{X}^{l+1} &= \mathbf{X}^l + \text{SW-MSA}\left(\text{LayerNorm}\left(\mathbf{X}^l\right)\right) \\
\mathbf{X}^{l+1} &= \hat{X}^{l+1} + \text{MLP}\left(\text{LayerNorm}\left(\hat{X}^{l+1}\right)\right). \qquad (1)
\end{aligned}$$

In layer l and layer $l+1$, the multihead attention is calculated using regular windows (denoted as W-MSA) and shifted windows (denoted as SW-MSA) successively. The shifted windows are asymmetrically divided from the original image, so that the boundary features are included in the shifted window-based attention. Layer normalization is applied before each MSA and each MLP. The attention function in them is

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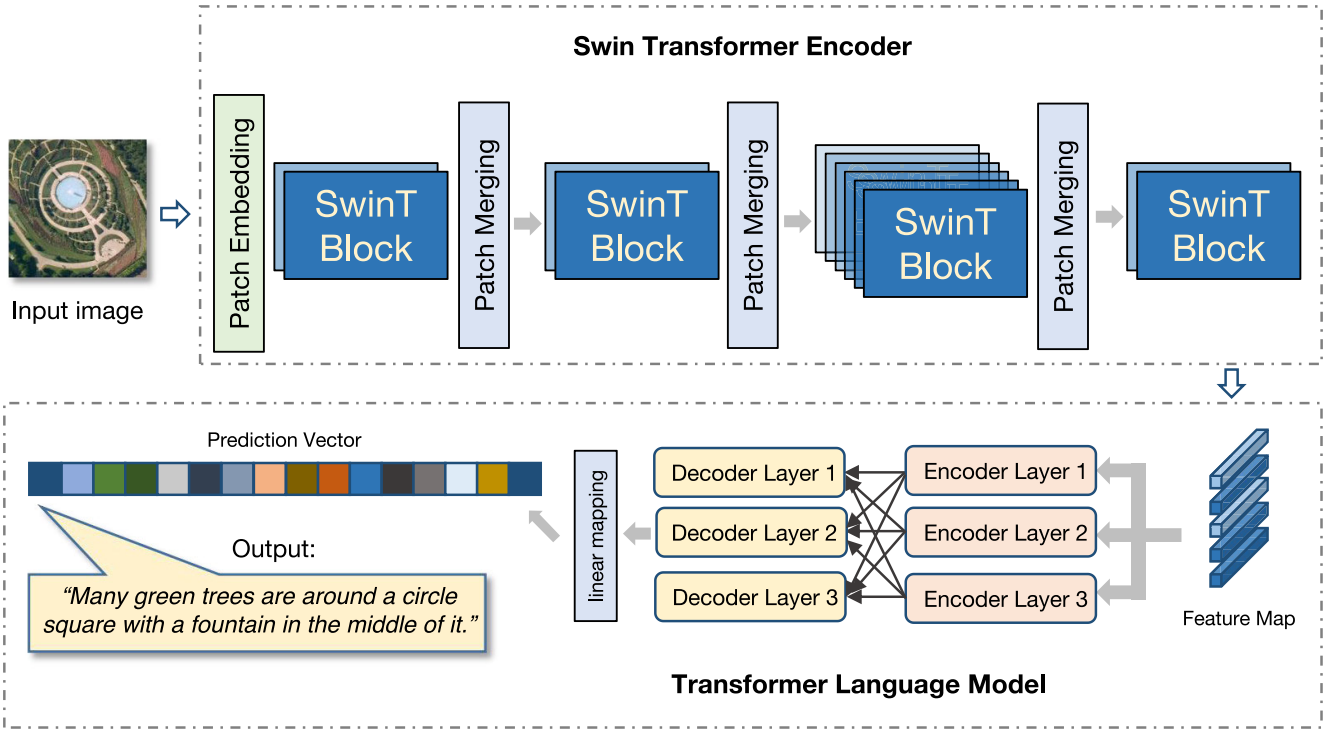


Fig. 1. Overall framework of the proposed method for RSIC. The feature maps of the input images are extracted by SwinT encoder, which consists of groups of SwinT blocks and patch operating modules. Then, the prediction sentences are calculated by Transformer decoder.

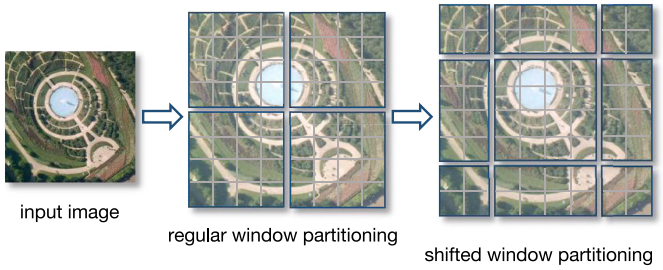


Fig. 2. Window partitioning scheme in patch partition module. The gray boxes are patches, and the blue boxes are local windows. In layer l, the image is divided into four local windows by regular window partitioning method, while in layer $l+1$, shifted window partitioning is adopted to generate nine new local windows, which contain boundary information of the regular windows.

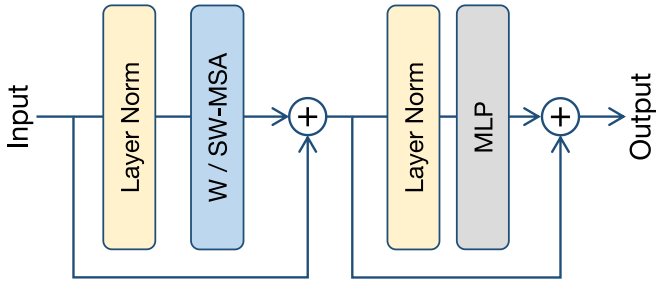


Fig. 3. Flowchart of SwinT block. The W/SW-MSA is regular window-based or shifted window-based MSA modules alternatively. The attention of the input feature is calculated after a layer normalization module by residual connection. Then, an MLP is added with another layer normalization module for the following feature transformation.

computed as follows:

$$\text{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^{\top}}{\sqrt{d}} + B\right)\mathbf{V} \tag{2}$$

$$\mathbf{Q} = X\mathbf{P}_Q, \mathbf{K} = X\mathbf{P}_K, \mathbf{V} = X\mathbf{P}_V \tag{3}$$

where $\mathbf{Q}$, $\mathbf{K}$, $\mathbf{V} \in \mathbb{R}^{M^2 \times d}$ represent the query, key, and value matrices, respectively. $\mathbf{P}_Q$, $\mathbf{P}_K$, and $\mathbf{P}_V$ denote the projection matrices shared among the windows. B is the learnable bias matrix. d indicates the dimension of the input. The number of patches in a window is $M \times M$. We utilize $M = 4$ in our implementation. The flowchart of a SwinT block is illustrated in Fig. 3.

C. Swin Transformer Encoder

The scale of the ground objects is spread over a very broad range. Inspired by the hierarchical ViT [58] in image classification and dense prediction tasks, we adopt groups of multiscale SwinT blocks as our image encoder. For the adjacent layers, W-MSA and SW-MSA are employed successively to establish cross-window connections. For the input RGB image with the size of $H \times W \times 3$, the patch partition module first reshapes it into the size of $(H/4) \times (W/4) \times 48$ by partitioning the 16 local windows. Patch merging layers concatenate the adjacent patches to reduce the feature resolution. The SwinT blocks with the aforementioned MSAs are applied on each local window. This process is repeated three times in SwinT encoder, in order to generate the hierarchical feature maps with the size of $(H/8) \times (W/8)$, $(H/16) \times (W/16)$, and $(H/32) \times (W/32)$, respectively, which could overcome the multiscale ambiguity in the remote sensing images. The group numbers are $\{2, 2, 6, 2\}$ by default.

D. Transformer Language Model

We adopt a multilayer TLM to better establish the relationship of features and generate a smooth sentence to describe

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TABLE I
ABLATION COMPARISON OF DIFFERENT BACKBONE NETWORKS ON THE THREE RSIC DATASETS. THE RESULTS ARE REPORTED AS PERCENTAGES (%)

Dataset	Encoder	Decoder	BLEU-1	BLEU-2	BLEU-3	BLEU-4	METEOR	ROUGE_L	CIDEr
UCM-Captions	ResNet50	LTLM	62.74	41.80	28.96	20.72	25.84	56.29	125.58
	ResNet101	LTLM	60.39	42.79	30.47	22.49	27.09	57.52	135.84
	ResNet50	TLM	84.06	72.23	63.08	55.38	40.04	75.97	275.72
	SwinT	LTLM	71.35	55.93	44.74	36.54	31.97	64.26	172.55
	SwinT	TLM	81.56	70.91	62.20	54.69	39.78	74.42	247.42
Sydney-Captions	ResNet50	LTLM	62.25	45.34	34.86	27.56	27.01	55.24	111.44
	ResNet101	LTLM	68.28	48.81	36.99	29.07	28.46	58.20	123.68
	ResNet50	TLM	71.72	54.12	41.54	32.32	29.98	60.71	140.03
	SwinT	LTLM	80.23	66.05	54.57	45.60	36.55	69.83	201.90
	SwinT	TLM	82.70	69.94	60.02	51.99	38.03	72.20	227.28
RSICD	ResNet50	LTLM	57.61	35.52	22.43	15.33	22.39	45.36	60.28
	ResNet101	LTLM	57.58	35.35	22.43	15.25	22.38	45.55	59.90
	ResNet50	TLM	61.06	39.52	26.48	18.66	24.23	49.29	77.05
	SwinT	LTLM	58.13	35.39	22.54	15.25	22.44	45.93	59.47
	SwinT	TLM	62.01	39.37	26.71	19.32	23.99	48.95	75.18

the image content. This attention-based TLM has a subencoder and subdecoder. The subencoder is built with a stack of three identical encoding layers. In each layer, MSA module is followed by a positionwise fully connected feedforward network (FFN), which is composed of two linear transformations and a ReLU activation function:

$$\mathrm{FFN}(x) = \max(0, xW_1 + b_1)W_2 + b_2 \tag{4}$$

where W_1 and W_2 are learnable weights. b_1 and b_2 are bias. The output features from the subencoder are then passed through linear projection layers to the subdecoder network, which consists of aggregated cross attention and FFN in each decoding layer. All the attentions are conducted in the form of residual connection. Finally, the subdecoder is connected with linear mapping layers to map the probability vectors to the words in the vocabulary.

IV. EXPERIMENTS AND DISCUSSIONS

In this section, we provide the detailed implementation and analysis of the proposed RSIC method. First, the remote sensing datasets and the evaluation metrics used in our experiments are explained. Following that, we conduct ablation study and comparison with the state-of-the-art image captioning approaches. The quantitative experimental results and visualization of the proposed method with deep analysis are illustrated in detail.

A. Experimental Details

1) Dataset: All the experiments in this article were done on UCM-Captions, Sydney-Captions, and RSICD datasets. For each high spatial resolution (HSR) image, five captioning sentences are given as references. Fig. 4 displays several image examples from the datasets.

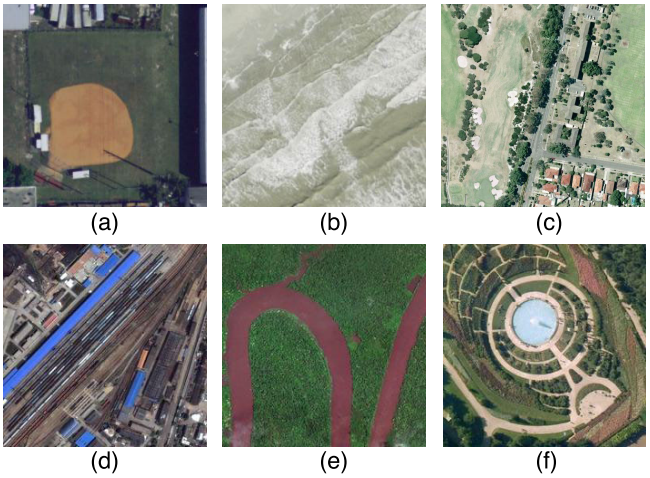


Fig. 4. Sample images in RSIC datasets. The corresponding reference sentences are as follows. (a) It is a small baseball diamond with sand and grass. (b) Waves come over again and again to the sand beach and throw up white foam. (c) Meadow with some green bushes and white bunkers on it while a residential area beside. (d) Some buildings are in two sides of a railway station with a long blue building. (e) Many green trees are in two sides of two curved khaki rivers. (f) Many green trees are around a circle square with a fountain in the middle of it.

1) *UCM-Captions Dataset* provided by [9] is based on the remote sensing scene classification dataset UC Merced Land Use [64] from Google Earth. There are 2100 images in total divided by 21 scene categories. Each of them contains 100 images. The image size is 256×256 pixels. The spatial resolution is about 0.3 m.
2) *Sydney-Captions Dataset* is also provided by [9]. The images are from the remote sensing scene classification dataset Sydney [65] from Google Earth. There are 613 images in total divided by seven scene categories

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TABLE II

PERFORMANCE COMPARISON WITH REPRESENTATIVE STATE-OF-THE-ART METHODS ON THE UCM-CAPTIONS DATASET. THE RESULTS ARE REPORTED AS PERCENTAGES (%). THE BEST SCORES AND THE SECOND BEST SCORES ARE DISPLAYED IN BLUE AND REDRED, RESPECTIVELY

Method	BLEU-1	BLEU-2	BLEU-3	BLEU-4	METEOR	ROUGE_L	CIDEr
SIFT+RNN [42]	57.30	44.35	37.96	33.88	24.61	53.34	170.33
BoW+RNN [43]	41.07	22.49	14.52	10.95	10.98	34.39	30.72
BoW+LSTM [43]	39.11	18.77	10.89	7.06	13.15	33.14	17.81
FV+LSTM [44]	58.97	46.68	40.80	36.83	26.98	55.95	184.38
VLAD+RNN [45]	63.11	51.93	46.06	42.09	29.71	58.78	200.66
DeViSE [59]	36.97	17.40	9.77	5.98	9.77	29.73	9.70
CSMLF (with CCA) [60]	38.74	21.45	12.53	9.15	9.54	35.99	37.03
CSMLF (fine-tuned model) [60]	36.71	14.85	7.63	5.05	9.44	29.86	13.51
Sound-A-F [61]	78.28	**72.76**	**67.59**	53.33	38.03	68.64	**290.57**
Sound-F-F [61]	72.07	65.32	59.37	**54.71**	34.15	62.96	249.62
IllustratedGPT [62]	25.16	6.62	1.22	0.27	23.56	17.33	-
PromptCap [33]	27.26	9.92	2.95	0.66	27.15	19.52	-
The proposed TrTr-CMR	**81.56**	70.91	62.20	54.69	**39.78**	**74.42**	247.42

TABLE III

PERFORMANCE COMPARISON WITH REPRESENTATIVE STATE-OF-THE-ART METHODS ON THE SYDNEY-CAPTIONS DATASET. THE RESULTS ARE REPORTED AS PERCENTAGES (%). THE BEST SCORES AND THE SECOND BEST SCORES ARE DISPLAYED IN BLUE AND REDRED, RESPECTIVELY

Method	BLEU-1	BLEU-2	BLEU-3	BLEU-4	METEOR	ROUGE_L	CIDEr
SIFT+RNN [42]	58.89	48.18	42.68	38.94	26.16	53.92	121.72
FV+LSTM [44]	63.31	53.32	47.35	43.03	29.67	57.94	147.61
VLAD+RNN [45]	56.58	45.14	38.07	32.79	26.72	52.71	93.72
DeViSE [59]	64.15	51.53	43.47	38.08	27.04	56.63	139.16
Multi-modal GRU [63]	69.64	60.92	52.39	44.21	31.12	59.17	171.55
CSMLF (with CCA) [60]	44.41	33.69	28.15	24.08	15.76	40.18	93.78
CSMLF (fine-tuned model) [60]	59.98	45.83	38.69	34.33	24.75	50.18	75.55
Sound-A-F [61]	59.82	51.75	43.84	36.73	25.29	51.46	134.18
Sound-F-F [61]	57.07	49.22	40.77	33.49	25.03	48.82	119.73
IllustratedGPT [62]	23.65	8.93	2.49	0.04	25.59	18.26	-
PromptCap [33]	25.69	7.83	5.03	0.95	27.90	17.96	-
The proposed TrTr-CMR	**82.70**	**69.94**	**60.02**	**51.99**	**38.03**	**72.20**	**227.28**

in the dataset. Each of them is 500 × 500 pixels. The spatial resolution is about 0.5 m.

3) *RSICD* is a large-scale aerial dataset provided in [10]. The remote sensing images in RSICD are collected from different sources (Baidu Map, MapABC, Google Earth, and Tianditu). The scene categories include airport, baseball field, mountain, residential area, square, viaduct, and so on. The images are all 224×224 pixels in size but with different spatial resolutions. In this dataset, there are 10 921 images with high intraclass diversity and low interclass dissimilarity.

2) Evaluation Metrics: The metrics in this article include BLEU [66], METEOR [67], ROUGE_L [68], and CIDEr [69].

*a) BLEU (**Bi**Lingual **E**valuation **U**nderstudy):* is a widely used evaluation for machine translation. It measures the co-occurrences of n ordered words (n-gram) between the reference sentence and the generated sentence. Denote r and c as the total length of the reference and generated sentence, respectively. We first compute the brevity penalty (BP) to make a decaying exponential in r/c

$$P_B = \begin{cases} 1 & \text{if } \ c > r \\ e^{1-r/c} & \text{otherwise.} \end{cases} \tag{5}$$

Then, the BLEU score is calculated by

$$\text{BLEU} = P_B \cdot \exp\left(\sum_{n=1}^{N} w_n \log p_n\right) \tag{6}$$

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TABLE IV

PERFORMANCE COMPARISON WITH REPRESENTATIVE STATE-OF-THE-ART METHODS ON THE RSICD DATASET. THE RESULTS ARE REPORTED AS PERCENTAGES (%). THE BEST SCORES AND THE SECOND BEST SCORES ARE DISPLAYED IN BLUE AND REDRED, RESPECTIVELY

Method	BLEU-1	BLEU-2	BLEU-3	BLEU-4	METEOR	ROUGE_L	CIDEr
SIFT+RNN [42]	47.65	28.28	19.61	14.56	18.81	39.91	78.82
BoW+RNN [43]	44.01	23.83	15.14	10.41	16.84	36.05	46.67
BoW+LSTM [43]	48.18	29.08	20.42	15.33	19.44	39.95	91.52
FV+LSTM [44]	43.42	24.53	16.34	11.75	17.12	38.18	65.31
DeViSE [59]	30.68	11.38	5.58	3.07	9.73	25.63	12.44
Multi-modal GRU [63]	42.56	29.99	22.91	17.98	19.41	37.97	124.82
CSMLF (with CCA) [60]	57.59	38.59	28.32	22.17	21.28	44.55	52.97
CSMLF (fine-tuned model) [60]	51.06	29.11	19.03	13.52	16.93	37.89	33.88
Sound-A-F [61]	58.47	**43.80**	**34.52**	**27.85**	24.45	46.16	**137.17**
Sound-F-F [61]	54.09	38.56	28.72	21.70	22.76	44.15	85.10
IllustratedGPT [62]	26.34	6.57	1.11	0.28	21.02	14.09	-
PromptCap [33]	27.07	7.75	2.09	0.49	**24.56**	16.22	-
The proposed TrTr-CMR	**62.01**	39.37	26.71	19.32	23.99	**48.95**	75.18

where p_n is the geometric average of the modified n-gram precision score based on clipped n-gram counts

$$p_n = \frac{\sum_{C\in\{\text{cand}\}}\sum_{n-\text{gram}\in C}\text{Count}_{\text{clip}}(n-\text{gram})}{\sum_{C\in\{\text{cand}\}}\sum_{n-\text{gram}\in C}\text{Count}(n-\text{gram})}. \tag{7}$$

According to [66], we use uniform weights $w_n = 1/N$. The value of n in this article is 1, 2, 3, and 4.

*b) METEOR (**M**etric for **E**valuation of **T**ranslation with **E**xplicit **OR**dering):* is a metric for machine translation based on explicit word-to-word matching to address the weakness in BLEU. First, unigram precision P and unigram recall R are calculated by

$$P = m/c,\ R = m/r \tag{8}$$

where m is the number of mapped unigrams in the generated sentence. c and r are the total number of unigrams in the generated and reference sentence, respectively. Then, the F value is calculated via a harmonic mean of unigram precision and recall

$$F_{\text{means}} = \frac{10PR}{R+9P}. \tag{9}$$

To think of the longer matches, a chunk penalty factor is introduced

$$P_C = 0.5 * \left(\frac{\#\text{chunks}}{\text{unigrams_matched}}\right)^3. \tag{10}$$

Finally, the METEOR score is computed through the following formula:

$$\text{METEOR} = (1-P_C)\cdot F_{\text{means}}. \tag{11}$$

*c) ROUGE (**R**ecall-**O**riented **U**nderstudy for **G**isting **E**valuation):* is designed for machine translation and image captioning tasks. ROUGE_L calculates the F-measure based on the length of the longest common subsequence (LCS). For the reference sentence Y of length r and generated sentence X of length c, the precision and recall are

$$P_{\text{lcs}} = \frac{\text{LCS}(X,Y)}{c} \tag{12}$$

$$R_{\text{lcs}} = \frac{\text{LCS}(X,Y)}{r}. \tag{13}$$

Then, the ROUGE_L score is calculated as follows:

$$\text{ROUGE_}L = \frac{\left(1+\beta^2\right)P_{\text{lcs}}R_{\text{lcs}}}{R_{\text{lcs}}+\beta^2 P_{\text{lcs}}}. \tag{14}$$

*d) CIDEr (**C**onsensus-based **I**mage **D**escription **E**valuation):* measures the term frequency-inverse document frequency (TF-IDF)-related cosine similarity between reference and generated sentence for all n-grams. In image captioning dataset $\boldsymbol{I}$, each test image I_i has m reference sentences $s_i = \{s_{i1}, s_{i2}, \ldots, s_{im}\}$ and a generated sentence c_i. The number of occurrences of each n-gram ω_k in s_{ij} and c_i is denoted as $h_k(s_{ij})$ and $h_k(c_i)$. Then, the TF-IDF of ω_k is

$$g_k\left(s_{ij}\right) = \frac{h_k\left(s_{ij}\right)}{\sum_{w_l\in\Omega} h_l\left(s_{ij}\right)}\log\left(\frac{|\boldsymbol{I}|}{\sum_{I_p\in I}\min\left(1,\sum_q h_k\left(s_{pq}\right)\right)}\right) \tag{15}$$

where Ω represents the set formed by all n-grams. The cosine similarity between the reference and generated sentence is calculated by

$$\text{CIDEr}_n(c_i,s_i) = \frac{1}{m}\sum_j \frac{\boldsymbol{g}^n(c_i)\cdot\boldsymbol{g}^n\left(s_{ij}\right)}{\|\boldsymbol{g}^n(c_i)\|\|\boldsymbol{g}^n\left(s_{ij}\right)\|} \tag{16}$$

where $\|\cdot\|$ is the magnitude of vector. $\boldsymbol{g}^n(\cdot)$ is the vector composed of $g_k(\cdot)$ of all the n-grams. The CIDEr score is a weighted sum of all the CIDEr_n

$$\text{CIDEr}(c_i,s_i) = \sum_{n=1}^{N} w_n\text{CIDEr}_n(c_i,s_i). \tag{17}$$

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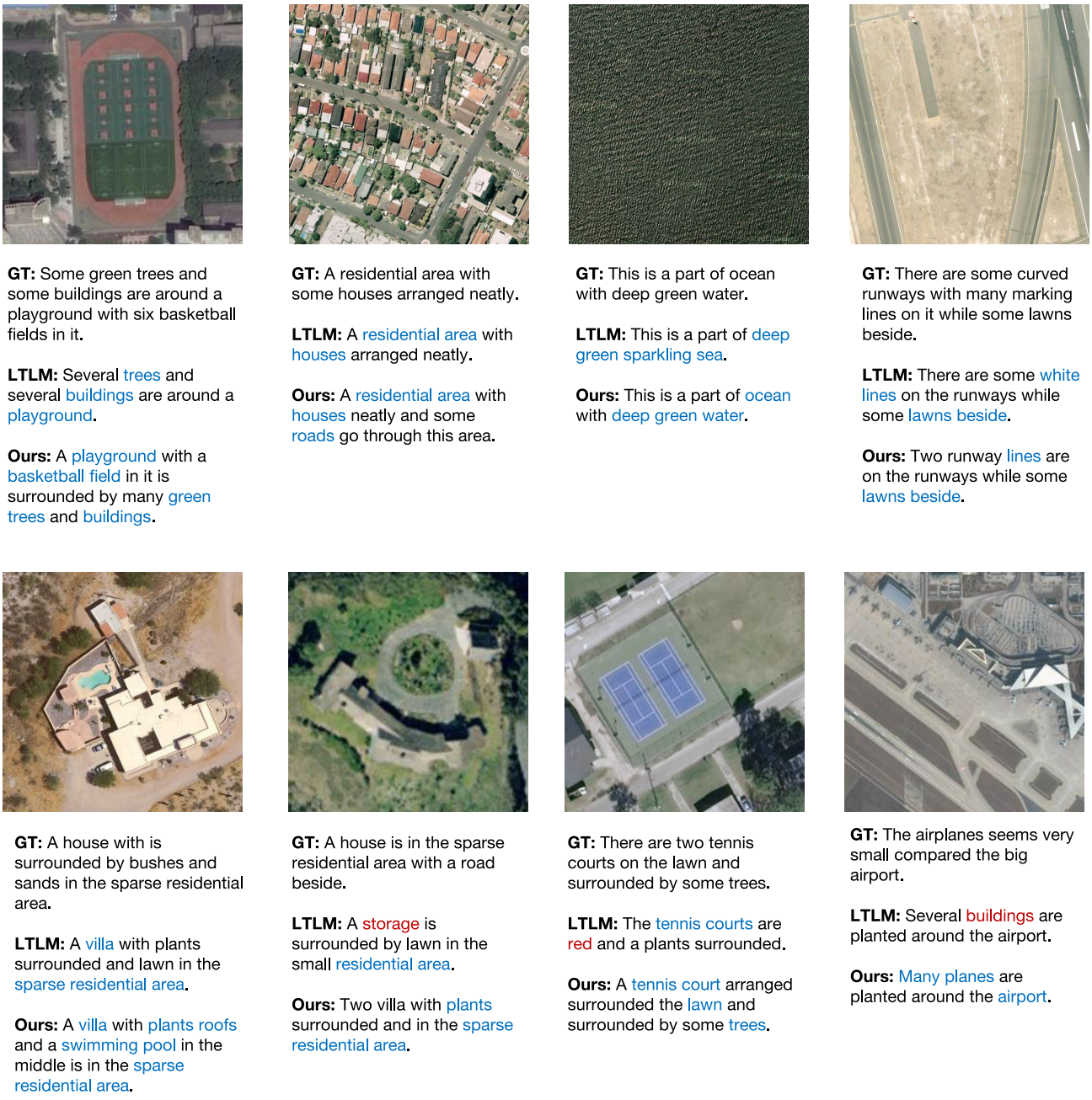


Fig. 5. Some representative examples of sentences generated by the proposed method and their corresponding reference sentences from the RSIC datasets. For each image, GT is the reference sentence, LTLM is the caption of TLM with LSTM, and Ours is the caption of our method. The accurately predicted words are highlighted in blue, and the discrepant words are in red color.

In general, $w_n = 1/N$ and $N = 4$.

The range of BLEU-1, BLEU-2, BLEU-3, BLEU-4, METEOR, and ROUGE scores is between 0 and 1. The larger score indicates the better performance of the image captioning model. We report the scores as percentage numbers for clarity in the following statement. All of the experimental results are calculated by Microsoft COCO Caption Toolkit.[1]

3) Model Settings: The input image is first resized to 224 × 224 pixels. The patch size is 4. The window size is set to 7 by default. In the MLPs, the activation function is GELU, and the hidden layers are 4. The batch size for training is 64. Learning rates of encoder and decoder are both 10^{-4}. AdamW [70] is utilized as the optimizer of the encoder and decoder. Besides, all of our experiments are implemented under Python 3.9 and PyTorch 2.3.0 + cu121 environment and run with an NVIDIA RTX 4090 GPU (24-GB memory).

[1]https://github.com/tylin/coco-caption

B. Ablation Study

To study the significant contributions of the Transformers in the proposed dual Transformer framework, we conduct extensive ablation experiments on UCM-Captions, Sydney-Captions, and RSICD datasets. The image encoders in the ablation experiments are ResNet50, ResNet101, or SwinT,

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Fig. 6. Some unsatisfying cases of sentences generated by the proposed method and their corresponding reference sentences from the RSIC datasets. For each image, GT is the reference sentence, and Ours is the caption of our method. The discrepant words are highlighted in red.

respectively. Moreover, the decoders are Transformer language model (denoted as TLM) or a Transformer with LSTM layers (denoted as LTLM). The experimental results are presented in Table I. For the RSICD dataset, ResNet50 and ResNet101 backbone networks obtain similar results, while for Sydney-Captions, ResNet101 gets higher scores than ResNet50. By replacing ResNet with the SwinT network, we achieve significant improvements for all three datasets in comparison with the classical CNN-based networks on all the evaluation metrics. In addition, when employing TLM instead of LTLM as the decoder, the BLEU, METEOR, ROUGE_L, and CIDEr scores all have shown varying degrees of improvement for all of the three datasets.

According to the above experimental results and analysis, we could draw the conclusion that the proposed dual Transformer architecture has distinct contributions to the RSIC model.

C. Quantitative Comparison With the SOTA Methods

In this experiment, the proposed method, termed TrTr-CMR, is compared with 11 well-known image captioning methods. They use different types of handcrafted or deep features and adopt different network architectures.

1) VLAD + RNN [45] utilizes the classic handcrafted feature, vectors of locally aggregated descriptor (VLAD), as the image encoder and a naive RNN as the text generator.
2) SIFT + RNN [42] utilizes the classic handcrafted feature, SIFT, as the image encoder and a naive RNN as the text generator.
3) BoW + RNN [43] utilizes BOV representation and RNN decoder model.
4) BoW + LSTM [43] utilizes BOV representation and LSTM decoder model.
5) FV + LSTM [44] applies FV representation and LSTM decoder model.
6) DeViSE [59] is a deep visual-semantic embedding model leveraging the semantic information among labels.
7) Multimodal GRU [63] is a deep visual-semantic embedding model based on multimodal gated recurrent units (GRUs).
8) CSMLF [60] is a collective semantic metric learning framework by mapping the visual and textual representation to the same hyperspace.
9) Sound active attention [61] framework introduces the sound-guided attention into the image captioning network to capture sound features to the sentence generation.
10) IllustratedGPT [62] is an illustrated Transformer image captioning framework based on ViT [31] and GPT-2 [71].
11) PromptCap [33] is a Prompt-guided vision-language model that takes a natural-language prompt to guide image description via GPT-3 in-context learning.

All of the compared methods are trained and tested under the same division rules. Part of the quantitative results of the compared method is borrowed from their original papers and the literary review [72]. The comparison results of the UCM-Captions dataset are shown in Table II. Due to the limited representation ability of the handcrafted features, SIFT, BoW, and FV approaches obtain relatively inferior scores of BLEU, METEOR, and ROUGE_L metrics. Benefits from the superior visual feature extraction ability of the attention mechanism, our method obtains the highest BLEU-1, METEOR, and ROUGE_L scores, while the sound active attention model achieves comparable performance on the other evaluation metrics. The comparison results of the Sydney-Captions dataset are shown in Table III. Our method achieves the highest scores and outperforms the compared methods by a large margin, including the popular GPT-based approaches, because their generation results depend on the prompt words, and their models are pretrained on large-scale vision-language datasets, which have a different corpus from our RSIC context. The comparison results of RSICD dataset are shown in Table IV. This is a challenging dataset because of the high intraclass diversity and low interclass dissimilarity in the images; thus, the models achieve lower scores than the first two datasets.

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Our method obtains the highest BLEU-1 and ROUGE_L (i.e., 62.01% and 48.95%). The sound active attention model achieves the highest BLEU-2, BLEU-3, BLEU-4, and CIDEr, and PromptCap gets the best METEOR score.

D. Qualitative Experimental Results

Furthermore, to visually evaluate the performance of image captioning, eight intuitive examples from the UCM-Captions, Sydney-Captions, and RSICD datasets are provided in Fig. 5. For each image, the reference sentence (GT) and the prediction sentence by LTLM and our method (Ours) are displayed under the image. The words highlighted in blue are the accurately predicted words according to the image content. It could be found that our method generates well-formed sentences for most of the images. For instance, in the first image, the proposed method depicts the playground and the basketball field in it and notices the green trees and buildings around it. In the second image, LTLM misses the road crossing the residential area, but our method describes it successfully. In the fifth image, there is a grand villa surrounded by bushes and sand. Our method depicts the swimming pool in the area, which is ignored by LTLM and the human-annotated reference sentence. In the sixth image, there is a circular lawn at the center of the area. However, LTLM regards it as a storage. Overall, the visualization results in Fig. 5 demonstrate that our approach has a good sense of describing detailed features of the image.

E. Limitations

Despite our method outperforming the start-of-the-art methods on the existing RSIC datasets, there are still several limitations that need to be improved. Fig. 6 displays four typical unsatisfying cases by the proposed method, where the discrepant words are highlighted in red. For example, in the first image, there are small blue-roofed houses at the corner of the baseball field, but our method misses it. In the second image, there is a huge highway viaduct stretching across the scenario. However, the prediction ignores the highway viaduct, and the bare land is falsely identified as a playground. This indicates that the model lacks prior knowledge of cityscape. Considering the third and fourth images, our method describes the tennis courts and the storage tanks, but makes a mistake about the number of them, which indicates the weakness of object detection or counting.

V. Conclusion and Future Directions

This article proposes a cross-modal reasoning framework for RSIC. To leverage the visual-linguistic reasoning potential of the Transformer, a dual Transformer structure is designed, which is composed of SwinT encoder and a TLM as a sentence generator. The experimental results on the existing public RSIC benchmark datasets UCM-Captions, Sydney-Captions, and RSICD demonstrate that our method shows a promising performance for RSIC task.

For future works, there are still some problems when the approach is applied in the wild. Due to the limited corpus collected from the training set, it is sometimes difficult to generate well-organized sentences for new scenes or objects. Therefore, further studies on transfer learning or few-shot learning are needed. More flexible embedding-based and learning-based evaluation metrics that conform to human-like expressions should be explored when ground-truth captions are not available. In addition, other interesting man–machine interactions for remote sensing images, such as VQA and RSVG, are still an open issue that needs to be solved.

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